

CAP 182 SS2

Part D

Executive Summary

The aim of this project is to support client decision making by manufacturing a system that can predict whether an e-commerce order is likely to be returned. Two machine-learning models were developed and tested using a public dataset, these were Logistic Regression and Random Forest classification. The results of the evaluations and comparison indicate that while Logistic regression provides a slightly stronger performance overall, both models performed only modestly better than random assignment. This implies that additional information on customers, products and transactions could prove very valuable and that more complex models suite the problem better.

Recommended Model

Based on the results of the model performances and comparison between them, logistic regression would be the recommended model of the two. Logistic regression achieved an accuracy of 56.58%, precision of 54.09%, recall of 56.27%, F1 of 55.16% and and ROC-AUC of 59.01%. Random forest, on the other hand, achieved an accuracy of 55.94%, precision of 54.60%, recall of 42.53%, F1 of 47.82% and an ROC-AUC of 57.68%.

The most notable difference between the model's performances is in recall and F1-score. Logistic regression managed to identify approximately 56% of actual returns compared to the 43% of random forest. This difference is crucial as failing to identify a likely return will prevent the business from taking adequate preventative actions. Logistic regression also achieved a higher ROC-AUC score, indicating its greater capability in discriminating between returned and non-returned orders.

Although logistic regression performed to a greater degree than random forest, its performance was still largely weak. It should not be considered a highly accurate automated decision making system, rather, it should be viewed as an initial predictive model that could support further investigation and targeted intervention.

Improving the Model

Focusing on the chosen logistic regression model, improvements could be made primarily through greater data and feature engineering. The public dataset used for these examples provides useful transaction variables, however, the low ROC-AUC score of 0.5901 indicates that valuable explanatory information could be missing from the dataset.

Historical customer information could prove very valuable to this project. This could include purchase frequency, historical return rates, total prior returns and time since previous purchase. Zhu et al. (2018) demonstrates the importance of purchase and return history and customer/product similarity when predicting product returns.

Interaction features could also be worth investigating. Cui, Rajagopalan and Ward (2020) found that models that incorporate detailed interaction effects can improve predictive performances on return volume. Interactions such as product category with discount, customer history with product type, and delivery performance with order characteristics could prove valuable if evaluated.

The model could be tuned using cross-validation and suitable regularisation. Varying values of logistic regression's regularisation parameter could be evaluated while probability thresholds could be adjusted in accordance to the business objective.

Lastly, not in regards to the logistic regression model specifically, other algorithms should be investigated in future experimentation. Mishra and Dutta (2024) compared Logistic regression, Random Forest and Gradient Boosting on e-commerce return predictions and found that Gradient Boosting achieved the strongest ROC-AUC score than the other models, implying that it may be a better model to test. This research also identifies order amount, product category and cash on delivery charge as very relevant variables.

Adapting the Model to Client's Data

The primary adaptation would be mapping the client's data fields to the variables used during model development. The preprocessing pipeline would then need to handle the client's actual missing values, categorical variables, numerical ranges and potential duplicates. New features could also be derived from variables included in the client's data.

An important requirement in this regard is prevention of data leakage. Only the information that is available when the prediction is made should be used. Other variables such as return reason, refund amount or other information generated after the return should not be included as predictors. The point of prediction needs to be explicitly defined, e.g: immediately after the order is placed.

The model would also need to be retrained and validated on the client's data rather than assuming that the performance on the current public dataset will transfer cleanly to the client's data environment. This is crucial as differences in customers, products, sellers, operating procedures as well as different available features can drastically affect model performance.

The model should also be retrained and validated on the client's historical data rather than assuming that performance on the public dataset will transfer directly to the client's environment. This is important because differences in customers, products, sellers, geography and operating procedures can affect model performance.

References

- Cui, H., Rajagopalan, S. and Ward, A.R. (2020) 'Predicting product return volume using machine learning methods', *European Journal of Operational Research*, 281(3), pp. 612–627. doi:10.1016/j.ejor.2019.05.046.
- Mishra, A. and Dutta, P. (2024) 'Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns', *Journal of Cleaner Production*, 477, 143802. doi:10.1016/j.jclepro.2024.143802.
- Zhu, Y., Li, J., He, J., Quanz, B.L. and Deshpande, A.A. (2018) 'A local algorithm for product return prediction in e-commerce', *Proceedings of the Twenty-Seventh International Joint Conference on Artificial Intelligence*, pp. 3718–3724. doi:10.24963/ijcai.2018/517.